Mealvana Nutrition Algorithm v3

Introduction

This algorithm calculates personalized nutrition targets for endurance athletes across six macros: carbohydrates, protein, fat, fiber, sodium, and hydration. It covers four workout phases: pre-workout (with three timing windows), during exercise, post-workout, and transitions (for brick workouts).

The algorithm is built primarily from guidelines published in the Mealvana blog ("How Mealvana Calculates Your Fueling"), which synthesizes evidence from 66+ peer-reviewed sports nutrition studies. Where the blog provides specific values or formulas, we use them directly. Where the blog mentions a variable but doesn't provide a formula (such as how intensity affects during-exercise carbs), we've designed our own approach and marked it clearly.

The core principle is that nutrition needs scale with three main factors: (1) workout duration determines the base carbohydrate band, (2) sport type determines the practical ceiling due to gastrointestinal tolerance differences, and (3) individual factors like gut training, sweat rate, and environmental conditions personalize the recommendations within those bounds.

For intensity, the blog states that "duration, intensity (TSS/IF), sport type, environmental conditions, personal tolerance, and training context all influence recommendations simultaneously" but doesn't specify the formula. We've implemented intensity as a "range slider" that positions the athlete within the duration-based carb band—higher intensity aims toward the upper end of the range, lower intensity toward the lower end.

Example Personas

Throughout this document, we calculate examples for three personas to illustrate how the formulas work across different athlete profiles.

Persona	Description	Weight	Gut Training	Sweat Rate	Sweat Sodium
Elena	Elite female runner, 28 years old, competitive marathoner	52 kg	high	medium	medium
Marcus	Male age-grouper, 42 years old, experienced triathlete	75 kg	moderate	medium	medium
Sarah	Female beginner, 35 years old, training for first half-marathon	82 kg	low	light	low

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Example Workouts

We use three different workouts to show how the algorithm adapts across duration, intensity, and sport type. Combined with three personas, this gives 9 example calculations per phase.

Workout A: Moderate 2-Hour Run

- **Duration:** 120 min
- **Sport:** run

- **Intensity:** 30% Z1-2, 50% Z3, 20% Z4-5
- **Temperature:** 22°C
- **Humidity:** 55%
- **Use case:** Typical weekend long run with tempo segments

Workout B: Easy 45-Minute Recovery Run

- **Duration:** 45 min
- **Sport:** run
- **Intensity:** 90% Z1-2, 10% Z3, 0% Z4-5
- **Temperature:** 18°C
- **Humidity:** 50%
- **Use case:** Recovery day, candidate for fasted training

Workout C: Long 4-Hour Bike Ride

- **Duration:** 240 min
- **Sport:** bike
- **Intensity:** 40% Z1-2, 45% Z3, 15% Z4-5
- **Temperature:** 28°C
- **Humidity:** 70%
- **Use case:** Century ride prep, hot conditions

Input Variables

Variable	Unit	Description
`weight`	kg	Athlete's body weight
`duration`	min	Total workout duration
`sport`	enum	`run`, `bike`, or `swim`
`pct_zone_low`	0-1	Fraction of workout spent in zones 1-2 (easy/recovery)
`pct_zone_mid`	0-1	Fraction of workout spent in zone 3 (tempo)
`pct_zone_high`	0-1	Fraction of workout spent in zones 4-5 (threshold/VO2max)
`temperature`	°C	Environmental temperature (default: 20)
`humidity`	%	Relative humidity (default: 60)
`gut_training`	enum	`low`, `moderate`, or `high` (default: moderate)
`sweat_rate`	enum	`light`, `medium`, or `heavy` (default: medium)
`sweat_sodium`	enum	`low`, `medium`, or `high` (default: medium)
`has_full_meal`	bool	Whether athlete will eat a full meal 3-4h before (default: true)
`has_snack`	bool	Whether athlete will eat a snack 1-2h before (default: true)
`has_topup`	bool	Whether athlete will have a top-up 30-60 min before (default: true)
`is_fasted`	bool	Whether athlete is training fasted (default: false)

Constraints:

- `pct_zone_low + pct_zone_mid + pct_zone_high = 1.0`
- `is_fasted = true` implies `has_full_meal = false AND has_snack = false AND has_topup = false`

Lookup Tables

These tables are referenced throughout the algorithm. Use them to look up values based on input parameters.

TABLE 1: Duration → Carb Band (g/hr) — DURING EXERCISE

Used in Phase 2 (During Exercise) to determine base carbohydrate targets.

Duration	carb_band_low	carb_band_high	Source
< 60 min	0	30	Blog: "0-30g total, mouth rinse sufficient"
60-90 min	30	60	Blog: "30-60 g/hr, maintain blood glucose"
90 min - 2.5 hr	45	60	Blog: "45-60 g/hr, delay glycogen depletion"
2.5 - 4 hr	60	90	Blog: "60-90 g/hr, multiple transportable carbs"
> 4 hr	80	100	Blog: "80-100+ g/hr, maximum sustainable"

TABLE 2: Sport → Carb Ceiling (g/hr) — DURING EXERCISE

Used in Phase 2 to cap carbohydrate intake based on GI tolerance by sport.

Sport	carb_ceiling	Source
run	70	Blog: "Running 50-70 g/hr, GI jostling"
bike	120	Blog: "Cycling 80-120 g/hr, stable platform"
swim	0	Blog: "Swimming pre/post only, no during-activity"

TABLE 3: Gut Training → Absorption Cap (g/hr)

Used in Phase 2 to cap carbohydrate intake based on trained gut capacity.

gut_training	absorption_cap	Source
low	40	MEALVANA: Conservative for untrained gut
moderate	60	Blog: "60 g/hr single transporter ceiling"
high	100	Blog: "90-120 g/hr with trained gut + dual carbs"

TABLE 4: Pre-Workout Timing → Carb Target (g/kg) — CUMULATIVE WINDOWS

Used in Phase 1 (Pre-Workout). **These windows are CUMULATIVE** — athletes may eat at multiple windows, and targets are additive. Each window uses its standard blog-specified range regardless of which other windows are used.

Window	Time Before	carb_per_kg_low	carb_per_kg_high	Source
Full Meal	3-4 hours	1.0	4.0	Blog: "1-4 g/kg, full gastric emptying"
Snack	1-2 hours	1.0	2.0	Blog: "1-2 g/kg, snack window"
Top-Up	30-60 min	0.5	1.0	Blog: "0.5-1 g/kg, simple carbs only"
Emergency	< 30 min	0.0	0.5	Blog: "Emergency only, 25-50g"

Cumulative Pre-Workout Carbohydrate Considerations

When all three windows are used, cumulative carbs can theoretically reach 2.5-7 g/kg. This raises important questions about whether such high totals are appropriate.

What the Research Says:

The ISSN Position Stand on Nutrient Timing recommends "consuming carbohydrate quantities of 1 to 4 g/kg body mass in the 1 to 4 hours before exercise" for events lasting >60 minutes (Kerksick et al., 2017). Critically, this recommendation assumes a **single feeding window**, not stacked meals. No peer-reviewed research directly validates cumulative intake across multiple windows exceeding 4 g/kg.

Elite athlete practices support more conservative totals. Marathon and Ironman athletes typically consume 2-3 g/kg total pre-race: a substantial breakfast (1.5-2 g/kg) plus a small top-off (0.3-0.5 g/kg). Even aggressive carb-loading protocols (10-12 g/kg/day) spread intake across 24-48 hours, not concentrated in a 4-hour pre-exercise window (Burke et al., 2011).

Additional Concerns:

The 30-60 minute "top-up" window carries reactive hypoglycemia risk. Research shows hypoglycemic events peak when carbs are consumed 30-90 minutes before exercise, as insulin response coincides with exercise onset (Jeukendrup, 2017). Athletes consuming large amounts in this window without gut training may experience blood glucose crashes.

Our Approach:

This algorithm provides each window's blog-specified range without enforcing a cumulative cap, allowing flexibility for different athlete needs. However, we recommend athletes **target 2-4 g/kg total** across all windows for most training and racing scenarios, reserving higher intakes (4-5 g/kg) only for ultra-endurance events (>4 hours) with practiced gut training. Athletes should build toward higher totals gradually over training cycles rather than attempting maximum intake on race day.

Sources:

- Kerksick et al. (2017). ISSN Position Stand: Nutrient Timing. JISSN. https://pmc.ncbi.nlm.nih.gov/articles/PMC5596471/

- Burke et al. (2011). Carbohydrates for training and competition. J Sports Sci.
- Jeukendrup (2017). Training the gut for athletes. Sports Med.

TABLE 5: Sweat Rate → Base Rate (L/hr)

Used in Phase 2 for hydration and sodium calculations.

sweat_rate	sweat_base	Source
light	0.75	Baker et al. 2017: lower quartile
medium	1.25	Baker et al. 2017: median
heavy	2.0	Baker et al. 2017: upper quartile

TABLE 6: Sweat Sodium → Concentration (mg/L)

Used in Phase 2 for sodium loss calculations.

sweat_sodium	sodium_concentration	Source
low	550	Baker et al. 2017: ~25th percentile
medium	925	Baker et al. 2017: ~50th percentile
high	1150	Baker et al. 2017: ~75th percentile

Derived Values

```
duration_hours = duration / 60

intensity_position = (pct_zone_low × 0) + (pct_zone_mid × 0.5) + (pct_zone_high × 1.0)

env_sweat_multiplier = 1.0 + max(0, (temperature − 20) × 0.04)

actual_sweat_rate = sweat_base × env_sweat_multiplier
```

Derived values for all workouts:

Workout	duration_hours	intensity_position	env_sweat_multiplier
A (2hr run)	2.0	0.45	1.08
B (45min easy)	0.75	0.05	0.92
C (4hr bike)	4.0	0.375	1.32

Actual sweat rates by persona and workout:

Persona	Sweat Base	Workout A	Workout B	Workout C
Elena	1.25 L/hr	1.35 L/hr	1.15 L/hr	1.65 L/hr

Marcus	1.25 L/hr	1.35 L/hr	1.15 L/hr	1.65 L/hr
Sarah	0.75 L/hr	0.81 L/hr	0.69 L/hr	0.99 L/hr

Phase 1: Pre-Workout

How Pre-Workout Windows Work

Pre-workout nutrition consists of three timing windows that are **CUMULATIVE** when used together. Athletes may use one, two, or all three windows depending on their schedule.

The Three Windows (from blog):

Window	Timing	Carbs (g/kg)	Fat	Fiber	Notes
Full Meal	3-4 hr before	1-4	0.3-0.5 g/kg	Moderate	Full gastric emptying
Snack	1-2 hr before	1-2	<10g	Low	Snack window
Top-Up	30-60 min before	0.5-1	0g	0g	Simple carbs only

Cumulative Scenarios:

Scenario	Windows Used	Calculation
Full Fuel	Meal + Snack + Top-up	Standard calculations for each window
No Full Meal	Snack + Top-up only	Standard snack + standard top-up
Top-Up Only	Top-up only	Standard top-up calculation
Fasted	None	See Fasted Training Guidelines

No Adjustment Factors: Each window uses its standalone blog-specified range regardless of which other windows are used. The blog does not specify adjustment factors for skipped windows, and no research validates specific multipliers. Athletes naturally self-adjust based on hunger and experience.

Design Decision: We initially explored 25% and 50% adjustment factors for skipped windows but removed them due to lack of research support. See "Cumulative Pre-Workout Carbohydrate Considerations" above for guidance on total intake limits.

Sport-Specific Pre-Workout Adjustments

Sport	Pre-Workout Buffer	Adjustment	Source
Running	3-4 hours minimum	Standard formulas	Blog: "GI jostling, vertical impact"
Cycling	2-3 hours acceptable	Can eat closer to start	Blog: "Stable platform, no impact"
Swimming	2-3 hours minimum	Reduce fat/fiber more	Blog: "Horizontal position"

Cycling adjustment: Athletes can use the 1-2 hour window even for high-carb intake (up to 2 g/kg) because cycling's stable position allows faster gastric emptying.

Swimming adjustment: Prioritize easily digestible carbs; the horizontal position can cause discomfort with heavy meals.

Fasted Training Guidelines

Fasted training can enhance metabolic adaptations (fat oxidation, mitochondrial biogenesis) when strategically implemented, but must be limited to appropriate workout types.

Eligibility by Workout Type (from blog):

Workout Type	Suitability	Duration Limit	Rationale
Easy/Recovery (Z1-2)	✅ Excellent	<60 min	Fat oxidation dominant
Moderate steady-state (Z3)	⚠️ Acceptable	<75 min	Marginal, glycogen starting to deplete
Long workouts (any zone)	❌ Not recommended	>90 min	Glycogen depletion problematic
Tempo/Threshold (Z4)	❌ Not recommended	Any	High-quality work impaired
Intervals/VO2max (Z5)	❌ Avoid	Any	Glycolytic work severely impaired

Justification: Blog table "Fasted Training by Intensity" provides these guidelines. Research from Van Proeyen et al. (2010) and Impey et al. (2018) confirms fasted training benefits are limited to low-intensity work.

Algorithm Check for Fasted Eligibility:

```
fasted_eligible:
    if pct_zone_high > 0.1:      false   # Any significant Z4-5 work
    if pct_zone_mid > 0.3:       false   # >30% tempo work
    if duration > 75:            false   # >75 minutes
    if sport = swim:             false   # Swimming has unique demands
    else:                        true

fasted_warning:
    if duration > 60 AND fasted_eligible: "Consider light fueling for sessions >60
min"
```

If `is_fasted = true` AND `fasted_eligible = false`: Show warning: "This workout is not suitable for fasted training. High intensity or long duration workouts require pre-workout nutrition for performance and safety."

Post-Fasted Workout Priority: When training fasted, post-workout nutrition becomes MORE critical:

- Consume carbs within 30 minutes (1.2 g/kg instead of 1.0 g/kg)

- Protein within 30 minutes (0.35 g/kg instead of 0.3 g/kg)

Window A: Full Meal (3-4 hours before)

This window allows complete gastric emptying. Athletes can eat a full meal with carbs, protein, fat, and moderate fiber.

Carbs (g)

Formula:

```
meal_carb_grams = weight × carb_per_kg
```

Chain:

```
carb_per_kg = (pct_zone_low × 0.75) + (pct_zone_mid × 1.5) + (pct_zone_high × 1.75)

meal_carb_target = weight × carb_per_kg
meal_carb_low = weight × 1.0
meal_carb_high = weight × 4.0
```

Justification: Blog table "Pre-Workout Timing Windows" specifies 1-4 g/kg for 3-4 hours before. Blog table "Pre-Workout Carbs by Intensity Zone" specifies Z1-2: 0.5-1 g/kg, Z3-4: 1-2 g/kg, Z4-5: 1.5-2 g/kg. We use midpoints weighted by zone percentages.

Examples (3 personas × 3 workouts = 9 combinations):

Persona	Workout A (2hr run)	Workout B (45min easy)	Workout C (4hr bike)
Elena (52 kg)	**66g** (1.28 g/kg)	**41g** (0.78 g/kg)	**61g** (1.17 g/kg)
Marcus (75 kg)	**96g** (1.28 g/kg)	**59g** (0.78 g/kg)	**88g** (1.17 g/kg)
Sarah (82 kg)	**105g** (1.28 g/kg)	**64g** (0.78 g/kg)	**96g** (1.17 g/kg)

Workout B has low carb_per_kg (0.78) because 90% Z1-2 intensity.

Protein (g)

Formula:

```
meal_protein_grams = weight × 0.25
```

Justification: Blog table shows protein is part of the full meal window. Values based on typical pre-workout meal composition (0.2-0.35 g/kg).

Examples:

Persona	All Workouts
Elena (52 kg)	**13g**
Marcus (75 kg)	**19g**
Sarah (82 kg)	**21g**

Protein does not vary by workout intensity.

Fat (g)

Formula:

```
meal_fat_grams = weight × 0.4
```

Justification: Blog table "Pre-Workout Timing Windows" specifies 0.3-0.5 g/kg fat for 3-4 hours before.

Examples:

Persona	All Workouts
Elena (52 kg)	**21g**
Marcus (75 kg)	**30g**
Sarah (82 kg)	**33g**

Fiber (g)

Formula:

```
meal_fiber_grams = 5
```

Justification: Blog table specifies "Moderate" fiber for 3-4 hours before. We interpret moderate as 3-8g.

Examples: All personas, all workouts: **5g** (range: 3-8g)

Sodium (mg)

Formula:

```
meal_sodium_mg = sodium_base + env_sodium_bump
```

Chain:

```
sodium_base:
    if sweat_sodium = low:    300
    if sweat_sodium = medium: 450
    if sweat_sodium = high:   600

env_sodium_bump:
    if temperature > 25: 150
    if temperature > 20: 75
    else:                0
```

Justification: Blog cites Sims et al. 2007 recommending sodium loading before exercise, particularly in hot conditions.

Examples:

Persona	Workout A (22°C)	Workout B (18°C)	Workout C (28°C)

Elena (medium)	**525mg**	**450mg**	**600mg**
Marcus (medium)	**525mg**	**450mg**	**600mg**
Sarah (low)	**375mg**	**300mg**	**450mg**

Hydration (mL)

Formula:

```
meal_hydration_ml = weight × hydration_per_kg
```

Chain:

```
hydration_per_kg = 6
if temperature > 25: hydration_per_kg = 7
```

Justification: Blog recommends "prehydrate with beverages several hours before activity." Standard guidance is 5-7 mL/kg.

Examples:

Persona	Workout A (22°C)	Workout B (18°C)	Workout C (28°C)
Elena (52 kg)	**312mL**	**312mL**	**364mL**
Marcus (75 kg)	**450mL**	**450mL**	**525mL**
Sarah (82 kg)	**492mL**	**492mL**	**574mL**

Window B: Snack (1-2 hours before)

This window is for a light snack. Reduce fat and fiber to allow faster digestion.

Carbs (g)

Formula:

```
snack_carb_grams = weight × snack_carb_per_kg
```

Chain:

```
snack_carb_per_kg = (pct_zone_low × 0.75) + (pct_zone_mid × 1.25) + (pct_zone_high ×
1.5)

snack_carb_target = weight × snack_carb_per_kg
snack_carb_low = weight × 1.0
snack_carb_high = weight × 2.0
```

Justification: Blog table "Pre-Workout Timing Windows" specifies 1-2 g/kg for 1-2 hours before. Intensity affects the target within this range.

Examples:

Persona	Workout A (2hr run)	Workout B (45min easy)	Workout C (4hr bike)
Elena (52 kg)	**57g** (1.10 g/kg)	**41g** (0.78 g/kg)	**53g** (1.02 g/kg)
Marcus (75 kg)	**82g** (1.10 g/kg)	**59g** (0.78 g/kg)	**77g** (1.02 g/kg)
Sarah (82 kg)	**90g** (1.10 g/kg)	**64g** (0.78 g/kg)	**84g** (1.02 g/kg)

Protein (g)

Formula:

```
snack_protein_grams = weight × 0.15
```

Justification: Blog shows protein should be reduced in the snack window. Light protein (0.1-0.2 g/kg) is acceptable.

Examples:

Persona	All Workouts
Elena (52 kg)	**8g**
Marcus (75 kg)	**11g**
Sarah (82 kg)	**12g**

Fat (g)

Formula:

```
snack_fat_grams = 5
```

Justification: Blog table "Pre-Workout Timing Windows" specifies <10g fat for 1-2 hours before.

Examples: All personas, all workouts: **5g** (range: 0-10g)

Fiber (g)

Formula:

```
snack_fiber_grams = 1
```

Justification: Blog table specifies "Low" fiber for 1-2 hours before. We interpret low as 0-3g.

Examples: All personas, all workouts: **1g** (range: 0-3g)

Sodium (mg)

Formula:

```
snack_sodium_mg = (sodium_base × 0.5) + env_sodium_bump
```

Justification: Reduced sodium compared to full meal, but still beneficial for preloading.

Examples:

Persona	Workout A (22°C)	Workout B (18°C)	Workout C (28°C)
Elena (medium)	**300mg**	**225mg**	**375mg**
Marcus (medium)	**300mg**	**225mg**	**375mg**
Sarah (low)	**225mg**	**150mg**	**300mg**

Hydration (mL)

Formula:

```
snack_hydration_ml = weight × hydration_per_kg
```

Chain:

```
hydration_per_kg = 5
if temperature > 25: hydration_per_kg = 6
```

Justification: Slightly less hydration than full meal window, but still important for preloading.

Examples:

Persona	Workout A (22°C)	Workout B (18°C)	Workout C (28°C)
Elena (52 kg)	**260mL**	**260mL**	**312mL**
Marcus (75 kg)	**375mL**	**375mL**	**450mL**
Sarah (82 kg)	**410mL**	**410mL**	**492mL**

Window C: Top-Up (30-60 minutes before)

This window is for simple, fast-absorbing carbs only. No fat, no fiber, minimal protein.

Carbs (g)

Formula:

```
topup_carb_grams = weight × topup_carb_per_kg
```

Chain:

```
topup_carb_per_kg = (pct_zone_low × 0.5) + (pct_zone_mid × 0.75) + (pct_zone_high ×
1.0)

topup_carb_target = weight × topup_carb_per_kg
topup_carb_low = weight × 0.5
topup_carb_high = weight × 1.0
```

Justification: Blog table "Pre-Workout Timing Windows" specifies 0.5-1 g/kg for 30-60 minutes before.

Examples:

Persona	Workout A	Workout B	Workout C
Elena (52 kg)	**38g**	**27g**	**35g**
Marcus (75 kg)	**55g**	**39g**	**51g**
Sarah (82 kg)	**60g**	**43g**	**56g**

Protein (g)

Formula:

```
topup_protein_grams = 0
```

Justification: Blog table shows 0g protein for 30-60 min window. Protein slows digestion.

Examples: All personas, all workouts: **0g**

Fat (g)

Formula:

```
topup_fat_grams = 0
```

Justification: Blog table "Pre-Workout Timing Windows" specifies 0g fat for 30-60 minutes before.

Examples: All personas, all workouts: **0g**

Fiber (g)

Formula:

```
topup_fiber_grams = 0
```

Justification: Blog table specifies 0g fiber for 30-60 minutes before.

Examples: All personas, all workouts: **0g**

Sodium (mg)

Formula:

```
topup_sodium_mg = env_sodium_bump + 100
```

Justification: Final sodium top-up before exercise, especially important in warm conditions.

Examples:

Persona	Workout A (22°C)	Workout B (18°C)	Workout C (28°C)
All	**175mg**	**100mg**	**250mg**

Hydration (mL)

Formula:

```
topup_hydration_ml = topup_base + env_topup_bump
```

Chain:

```
topup_base = 200
env_topup_bump:
    if temperature > 25: 100
    else:                50
```

Justification: Final fluid top-up 30-60 minutes before workout. Blog mentions final hydration before activity.

Examples:

Persona	Workout A (22°C)	Workout B (18°C)	Workout C (28°C)
All	**250mL**	**250mL**	**300mL**

Pre-Workout Total Summary

For an athlete using all three windows, here are the cumulative pre-workout totals. See "Cumulative Pre-Workout Carbohydrate Considerations" in the Lookup Tables section for guidance on appropriate total intake.

Total Pre-Workout Carbs (Workout A - 2hr run, moderate intensity):

Persona	Full Meal	+ Snack	+ Top-up	**TOTAL**	**g/kg**
Elena (52 kg)	66g	57g	38g	**161g**	3.1
Marcus (75 kg)	96g	82g	55g	**233g**	3.1
Sarah (82 kg)	105g	90g	60g	**255g**	3.1

Total Pre-Workout Carbs (Workout B - 45min easy):

Persona	Full Meal	+ Snack	+ Top-up	**TOTAL**	**g/kg**
Elena (52 kg)	41g	41g	27g	**109g**	2.1
Marcus (75 kg)	59g	59g	39g	**157g**	2.1
Sarah (82 kg)	64g	64g	43g	**171g**	2.1

Total Pre-Workout Carbs (Workout C - 4hr bike, hot):

Persona	Full Meal	+ Snack	+ Top-up	**TOTAL**	**g/kg**
Elena (52 kg)	61g	53g	35g	**149g**	2.9

Marcus (75 kg)	88g	77g	51g	**216g**	2.9
Sarah (82 kg)	96g	84g	56g	**236g**	2.9

All totals fall within the recommended 2-4 g/kg range. Athletes should build toward these totals gradually and may use fewer windows based on schedule and tolerance.

Phase 2: During Exercise

Sport-Specific During-Exercise Rules

CRITICAL: The sport type fundamentally changes during-exercise nutrition:

Sport	Carbs	Hydration	Sodium	Source
Running	50-70 g/hr max	Standard	Standard	Blog: "GI jostling, vertical impact"
Cycling	80-120 g/hr max	Standard	Standard	Blog: "Stable platform, no impact"
Swimming	**0 g/hr** (pre/post only)	Lower (0.35-0.55 L/hr)	Lower	Blog: "No during-activity fueling"

Swimming Exception: Athletes cannot eat during swimming. All carbohydrate fueling must happen pre-workout and post-workout. However, for very long open-water swims (>2.5 hours), brief feeding stops at aid stations are possible—see Brick section.

Carbs (g/hr)

Formula:

```
if sport = swim:
    during_carb_rate = 0    # Cannot eat while swimming
else:
    during_carb_rate = min(carb_in_band, carb_ceiling, absorption_cap)
```

Chain (for run/bike):

```
carb_band_low, carb_band_high = lookup from TABLE 1 (Duration)
carb_ceiling = lookup from TABLE 2 (Sport)
absorption_cap = lookup from TABLE 3 (Gut Training)

carb_in_band = carb_band_low + intensity_position × (carb_band_high − carb_band_low)

during_carb_target = min(carb_in_band, carb_ceiling, absorption_cap)
during_carb_low = max(carb_band_low, during_carb_target − 10)
during_carb_high = min(carb_band_high, carb_ceiling, absorption_cap,
during_carb_target + 10)

during_carb_total = during_carb_target × duration_hours
```

Justification: Blog table "Duration-Based Carbohydrate Targets" (TABLE 1) provides the duration bands. Blog table "Sport-Specific Differences" (TABLE 2) provides the sport ceilings (run: 50-70 g/hr, bike: 80-120 g/hr, swim: pre/post only). Blog mentions gut training enables higher absorption (TABLE 3). **Intensity effect (MEALVANA):** Blog states intensity affects recommendations but provides no formula; we use intensity to position within the duration band.

Workout Parameters:

Workout	Duration Band	Carb Band	Sport Ceiling	Intensity Position
A (2hr run)	90min-2.5hr	45-60 g/hr	70 g/hr	0.45
B (45min run)	<60min	0-30 g/hr	70 g/hr	0.05
C (4hr bike)	>4hr	80-100 g/hr	120 g/hr	0.375
D (1hr swim)	60-90min	30-60 g/hr	**0 g/hr**	N/A

Examples (including swim):

Persona	Workout A (2hr run)	Workout B (45min run)	Workout C (4hr bike)	Workout D (1hr swim)
Elena (high gut)	**52 g/hr** (104g)	**2 g/hr** (1g)*	**88 g/hr** (350g)	**0 g/hr** (0g)
Marcus (mod gut)	**52 g/hr** (104g)	**2 g/hr** (1g)*	**60 g/hr** (240g)**	**0 g/hr** (0g)
Sarah (low gut)	**40 g/hr** (80g)**	**2 g/hr** (1g)*	**40 g/hr** (160g)**	**0 g/hr** (0g)

*Workout B: <60 min = mouth rinse sufficient; minimal carbs needed **Capped by absorption_cap (Sarah's low gut = 40g/hr, Marcus moderate = 60g/hr)

Protein (g/hr)

Formula:

```
during_protein_rate:
    if duration_hours > 4: 3 g/hr
    else:                  0 g/hr
```

Justification: Blog does not specify during-exercise protein. Research consensus is protein unnecessary except ultra-endurance (>4 hours).

Examples:

Persona	Workout A	Workout B	Workout C
All	**0 g/hr**	**0 g/hr**	**0 g/hr**

Note: Workout C is exactly 4 hours, so 0 g/hr. For workouts >4hr, use 3 g/hr.

Fat (g/hr)

Formula:

```
during_fat_rate:
    if duration_hours > 4: 2 g/hr
    else:                  0 g/hr
```

Justification: Blog does not specify during-exercise fat. Research consensus is fat unnecessary except ultra-endurance.

Examples:

Persona	Workout A	Workout B	Workout C
All	**0 g/hr**	**0 g/hr**	**0 g/hr**

Fiber (g/hr)

Formula:

```
during_fiber_rate = 0
```

Justification: Fiber during exercise causes GI distress.

Examples: All personas, all workouts: **0 g/hr**

Sodium (mg/hr)

Formula:

```
during_sodium_rate = actual_sweat_rate × sodium_concentration × 0.6
```

Chain:

```
sodium_loss_rate = actual_sweat_rate × sodium_concentration

during_sodium_target = sodium_loss_rate × 0.6
during_sodium_low = max(300, sodium_loss_rate × 0.5)
during_sodium_high = min(1200, sodium_loss_rate × 0.8)

during_sodium_total = during_sodium_target × duration_hours
```

Justification: Blog formula: "Sodium Loss (mg/hr) = Sweat Rate (L/hr) × Sodium Concentration (mg/L)". Blog states "Target 50-80% replacement during exercise."

Examples (mg/hr rate, then total for workout):

Persona	Workout A (22°C)	Workout B (18°C)	Workout C (28°C)
Elena (medium Na)	**749 mg/hr** (1498mg)	**638 mg/hr** (478mg)	**916 mg/hr** (3663mg)
Marcus (medium Na)	**749 mg/hr** (1498mg)	**638 mg/hr** (478mg)	**916 mg/hr** (3663mg)

Sarah (low Na)	**267 mg/hr** (534mg)	**228 mg/hr** (171mg)	**327 mg/hr** (1307mg)

Hydration (mL/hr)

Formula:

```
during_hydration_rate = actual_sweat_rate × 1000 × 0.75
```

Chain:

```
during_hydration_target = actual_sweat_rate × 1000 × 0.75
during_hydration_low = max(300, during_hydration_target × 0.8)
during_hydration_high = min(1200, during_hydration_target × 1.2)

during_hydration_total = during_hydration_target × duration_hours
```

Justification: Blog mentions matching 70-90% of sweat losses during exercise. We use 75% as the target.

Examples (mL/hr rate, then total for workout):

Persona	Workout A (22°C)	Workout B (18°C)	Workout C (28°C)
Elena	**1013 mL/hr** (2025mL)	**863 mL/hr** (647mL)	**1238 mL/hr** (4950mL)
Marcus	**1013 mL/hr** (2025mL)	**863 mL/hr** (647mL)	**1238 mL/hr** (4950mL)
Sarah	**608 mL/hr** (1215mL)	**518 mL/hr** (388mL)	**743 mL/hr** (2970mL)

During Exercise Summary Tables

Total During-Exercise Carbs:

Persona	Workout A (2hr)	Workout B (45min)	Workout C (4hr)
Elena	**104g**	**1g**	**350g**
Marcus	**104g**	**1g**	**240g**
Sarah	**80g**	**1g**	**160g**

Total During-Exercise Sodium:

Persona	Workout A (2hr)	Workout B (45min)	Workout C (4hr)
Elena	**1498mg**	**478mg**	**3663mg**
Marcus	**1498mg**	**478mg**	**3663mg**
Sarah	**534mg**	**171mg**	**1307mg**

Total During-Exercise Hydration:

Persona	Workout A (2hr)	Workout B (45min)	Workout C (4hr)

Elena	**2025mL**	**647mL**	**4950mL**
Marcus	**2025mL**	**647mL**	**4950mL**
Sarah	**1215mL**	**388mL**	**2970mL**

Phase 3: Post-Workout

Carbs (g)

Formula:

```
post_carb_grams = weight × post_carb_per_kg × fasted_multiplier
```

Chain:

```
post_carb_per_kg:
    if duration_hours > 2: 1.2
    else:                  1.0

fasted_multiplier:
    if is_fasted: 1.2   # Extra glycogen replenishment priority
    else:         1.0

post_carb_target = weight × post_carb_per_kg × fasted_multiplier
post_carb_low = post_carb_target × 0.85
post_carb_high = post_carb_target × 1.15
```

Justification: Blog table "Post-Workout Recovery" specifies "Glycogen resynthesis: 1.0-1.2 g/kg/hr." **Fasted multiplier (MEALVANA):** After fasted training, glycogen replenishment is more critical; research recommends prioritizing immediate carb intake.

Examples (9 combinations):

Persona	Workout A (2hr)	Workout B (45min)	Workout C (4hr)
Elena (52 kg)	**52g** (1.0 g/kg)	**52g** (1.0 g/kg)	**62g** (1.2 g/kg)
Marcus (75 kg)	**75g** (1.0 g/kg)	**75g** (1.0 g/kg)	**90g** (1.2 g/kg)
Sarah (82 kg)	**82g** (1.0 g/kg)	**82g** (1.0 g/kg)	**98g** (1.2 g/kg)

Workouts A & B use 1.0 g/kg (≤2hr). Workout C uses 1.2 g/kg (>2hr). If fasted: multiply by 1.2 (e.g., Elena fasted Workout B = 62g instead of 52g)

Protein (g)

Formula:

```
post_protein_grams = weight × post_protein_per_kg
```

Chain:

```
post_protein_per_kg:
    if is_fasted: 0.35   # Higher priority after fasted training
    else:         0.30

post_protein_target = weight × post_protein_per_kg
post_protein_low = max(20, post_protein_target × 0.85)
post_protein_high = min(40, post_protein_target × 1.15)
```

Justification: Blog table "Post-Workout Recovery" specifies "Muscle protein synthesis: 0.25-0.4 g/kg (20-40g)." **Fasted adjustment (MEALVANA):** Impey et al. (2018) recommends higher protein after low-carb training states.

Examples:

Persona	All Workouts (normal)	After Fasted Training
Elena (52 kg)	**16g**	**18g**
Marcus (75 kg)	**23g**	**26g**
Sarah (82 kg)	**25g**	**29g**

Protein does not vary by workout duration/intensity.

Fat (g)

Formula:

```
post_fat_grams = weight × 0.2
```

Justification: Blog does not specify post-workout fat. Research shows fat does not impair recovery.

Examples:

Persona	All Workouts
Elena (52 kg)	**10g**
Marcus (75 kg)	**15g**
Sarah (82 kg)	**16g**

Fiber (g)

Formula:

```
post_fiber_grams = 2
```

Justification: Blog does not specify post-workout fiber. Light fiber acceptable in recovery meals.

Examples: All personas, all workouts: **2g** (range: 0-5g)

Sodium (mg)

Formula:

```
post_sodium_mg = max(300, sodium_deficit × 0.5)
```

Chain:

```
total_sodium_lost = actual_sweat_rate × sodium_concentration × duration_hours
sodium_deficit = total_sodium_lost − during_sodium_total

post_sodium_target = max(300, sodium_deficit × 0.5)
post_sodium_low = post_sodium_target × 0.75
post_sodium_high = min(700, post_sodium_target × 1.25)
```

Justification: Blog table "Post-Workout Recovery" specifies rehydration "with 500-700 mg/L sodium."

Examples (9 combinations):

Persona	Workout A (22°C)	Workout B (18°C)	Workout C (28°C)
Elena	**500mg**	**300mg***	**700mg**
Marcus	**500mg**	**300mg***	**700mg**
Sarah	**300mg***	**300mg***	**435mg**

Floored at 300mg minimum. Capped at 700mg.

Hydration (mL)

Formula:

```
post_hydration_ml = max(500, hydration_deficit × 1.5)
```

Chain:

```
total_fluid_lost = actual_sweat_rate × duration_hours × 1000
hydration_deficit = total_fluid_lost − during_hydration_total

post_hydration_target = max(500, hydration_deficit × 1.5)
post_hydration_low = max(500, hydration_deficit × 1.25)
post_hydration_high = hydration_deficit × 1.75
```

Justification: Blog table "Post-Workout Recovery" specifies "Rehydration: 150% of fluid losses."

Examples (9 combinations):

Persona	Workout A (22°C)	Workout B (18°C)	Workout C (28°C)
Elena	**1013mL**	**500mL***	**2475mL**
Marcus	**1013mL**	**500mL***	**2475mL**
Sarah	**608mL**	**500mL***	**1485mL**

Floored at 500mL minimum.

Post-Workout Summary Tables

Total Post-Workout Carbs:

Persona	Workout A (2hr)	Workout B (45min)	Workout C (4hr)
Elena	**52g**	**52g**	**62g**
Marcus	**75g**	**75g**	**90g**
Sarah	**82g**	**82g**	**98g**

Total Post-Workout Protein:

Persona	All Workouts
Elena	**16g**
Marcus	**23g**
Sarah	**25g**

Phase 4: Transitions (Brick Workouts Only)

Transitions apply only to multi-sport brick workouts (e.g., triathlon). These are brief nutrition windows between segments.

T1 (After swim or first segment)

Formula:

```
t1_carbs = weight × 0.3
t1_sodium = 150 + (100 if temperature > 25 else 0)
t1_hydration = 200 + (100 if temperature > 25 else 0)
t1_protein = 0
t1_fat = 0
t1_fiber = 0
```

Justification (MEALVANA): Blog does not provide specific transition nutrition values. Based on need for rapid absorption in short window.

Examples by temperature:

Persona	Cool (18°C)	Moderate (22°C)	Hot (28°C)
Elena (52 kg)	16g / 150mg / 200mL	16g / 150mg / 200mL	16g / 250mg / 300mL
Marcus (75 kg)	23g / 150mg / 200mL	23g / 150mg / 200mL	23g / 250mg / 300mL
Sarah (82 kg)	25g / 150mg / 200mL	25g / 150mg / 200mL	25g / 250mg / 300mL

Format: carbs / sodium / hydration

T2 (After bike, before run)

Formula:

```
t2_carbs = weight × 0.35
t2_sodium = 100 + (75 if temperature > 25 else 0)
t2_hydration = 150 + (75 if temperature > 25 else 0)
t2_protein = 0
t2_fat = 0
t2_fiber = 0
```

Justification (MEALVANA): Blog does not provide specific transition values. T2 is slightly higher carbs—last opportunity before run.

Examples by temperature:

Persona	Cool (18°C)	Moderate (22°C)	Hot (28°C)
Elena (52 kg)	18g / 100mg / 150mL	18g / 100mg / 150mL	18g / 175mg / 225mL
Marcus (75 kg)	26g / 100mg / 150mL	26g / 100mg / 150mL	26g / 175mg / 225mL
Sarah (82 kg)	29g / 100mg / 150mL	29g / 100mg / 150mL	29g / 175mg / 225mL

Format: carbs / sodium / hydration

Post-Bike Run Penalty

Formula:

```
if current segment is run AND previous segment was bike:
    run_carb_ceiling = 70 × 0.75 = 52.5 g/hr
```

Justification: Blog states "20-30% reduction in run carb targets after bike leg." We use 25% midpoint.

Appendix: Complete Workout Summaries

Workout A: 2-Hour Moderate Run (22°C)

Phase	Elena (52kg, high gut)	Marcus (75kg, mod gut)	Sarah (82kg, low gut)
Pre-Meal	66g carb, 13g pro, 21g fat	96g carb, 19g pro, 30g fat	105g carb, 21g pro, 33g fat
Pre-Snack	57g carb, 8g pro, 5g fat	82g carb, 11g pro, 5g fat	90g carb, 12g pro, 5g fat
Pre-TopUp	38g carb, 0g pro, 0g fat	55g carb, 0g pro, 0g fat	60g carb, 0g pro, 0g fat
During	52g/hr (104g total)	52g/hr (104g total)	40g/hr (80g total)
During Na	749mg/hr (1498mg)	749mg/hr (1498mg)	267mg/hr (534mg)

During H2O	1013mL/hr (2025mL)	1013mL/hr (2025mL)	608mL/hr (1215mL)
Post	52g carb, 16g pro, 10g fat	75g carb, 23g pro, 15g fat	82g carb, 25g pro, 16g fat

Workout B: 45-Min Easy Run (18°C)

Phase	Elena (52kg)	Marcus (75kg)	Sarah (82kg)
Pre-Meal	41g carb, 13g pro, 21g fat	59g carb, 19g pro, 30g fat	64g carb, 21g pro, 33g fat
Pre-Snack	41g carb, 8g pro, 5g fat	59g carb, 11g pro, 5g fat	64g carb, 12g pro, 5g fat
Pre-TopUp	27g carb, 0g pro, 0g fat	39g carb, 0g pro, 0g fat	43g carb, 0g pro, 0g fat
During	~0g (mouth rinse OK)	~0g (mouth rinse OK)	~0g (mouth rinse OK)
During Na	478mg total	478mg total	171mg total
During H2O	647mL total	647mL total	388mL total
Post	52g carb, 16g pro, 10g fat	75g carb, 23g pro, 15g fat	82g carb, 25g pro, 16g fat

Note: Workout B is eligible for fasted training (easy intensity, <60 min). If fasted, skip pre-workout phases and prioritize post-workout nutrition.

Workout C: 4-Hour Bike Ride (28°C, Hot)

Phase	Elena (52kg, high gut)	Marcus (75kg, mod gut)	Sarah (82kg, low gut)
Pre-Meal	61g carb, 13g pro, 21g fat	88g carb, 19g pro, 30g fat	96g carb, 21g pro, 33g fat
Pre-Snack	53g carb, 8g pro, 5g fat	77g carb, 11g pro, 5g fat	84g carb, 12g pro, 5g fat
Pre-TopUp	35g carb, 0g pro, 0g fat	51g carb, 0g pro, 0g fat	56g carb, 0g pro, 0g fat
During	88g/hr (350g total)	60g/hr (240g total)	40g/hr (160g total)
During Na	916mg/hr (3663mg)	916mg/hr (3663mg)	327mg/hr (1307mg)
During H2O	1238mL/hr (4950mL)	1238mL/hr (4950mL)	743mL/hr (2970mL)
Post	62g carb, 16g pro, 10g fat	90g carb, 23g pro, 15g fat	98g carb, 25g pro, 16g fat

Note: Workout C demonstrates how gut training affects during-exercise targets. Elena (high) can absorb 88g/hr, Marcus (moderate) is capped at 60g/hr, Sarah (low) at 40g/hr.

Brick Workout Adjustments

Brick workouts (multi-sport sessions like triathlon) require special handling because:

1. Swimming segment has 0 carbs during (cannot eat)
2. Post-bike running has reduced GI tolerance
3. Transitions are opportunities for rapid fueling

Brick Calculation Principles

Use CUMULATIVE duration for the carb band lookup (TABLE 1).

Example: A brick with 30min swim + 90min bike + 30min run = 2.5 hours total duration → use the "2.5-4 hr" band (60-90 g/hr).

Justification: Blog mentions brick workouts use total duration. Current edge function confirms: "Uses same bands as running but based on cumulative duration."

Segment-Specific Adjustments

Segment	Carb Ceiling	Special Rules	Source
Swim	0 g/hr	Cannot eat while swimming	Blog: "pre/post only"
Bike	120 g/hr	Front-load nutrition here	Current edge function
Bike (before run)	120 g/hr + **20% bonus**	Extra loading before run	Current edge function: "+20% if followed by run"
Run (after bike)	**52.5 g/hr** (reduced)	25% reduction from normal 70 g/hr	Blog: "20-30% reduction in run carb targets after bike leg"

Brick Carb Allocation Formula

```
# Step 1: Get cumulative duration carb band
total_duration = swim_duration + bike_duration + run_duration
carb_band = lookup TABLE 1 using total_duration

# Step 2: Calculate carbs per segment
swim_carbs = 0  # Cannot eat

bike_carb_rate = min(carb_in_band, 120, absorption_cap)
if has_run_after_bike:
    bike_carb_rate = bike_carb_rate × 1.2  # Front-load 20% extra
bike_carbs = bike_carb_rate × bike_duration_hours

run_carb_ceiling = 70 × 0.75  # 52.5 g/hr post-bike penalty
run_carb_rate = min(carb_in_band, run_carb_ceiling, absorption_cap)
run_carbs = run_carb_rate × run_duration_hours

# Step 3: Add transition carbs
total_carbs = swim_carbs + t1_carbs + bike_carbs + t2_carbs + run_carbs
```

Brick Example: Olympic Triathlon

Workout: 30min swim + T1 + 60min bike + T2 + 30min run = 2 hours total **Athlete:** Marcus (75kg, moderate gut training, 60 g/hr absorption cap) **Conditions:** 25°C

Segment	Duration	Carb Rate	Carbs	Notes

Pre-Workout	-	-	96g	3-4h before meal
Swim	30 min	0 g/hr	0g	Cannot eat
T1	2 min	-	**23g**	0.3 g/kg quick carbs
Bike	60 min	60 g/hr × 1.2 = 72 g/hr	72g	Capped by gut, +20% for run
T2	2 min	-	**26g**	0.35 g/kg last chance
Run	30 min	52.5 g/hr	26g	Post-bike penalty applied
Post-Workout	-	-	75g	Recovery
TOTAL DURING	2 hr	-	**147g**	Swim + T1 + Bike + T2 + Run

Hydration and Sodium During Brick

Swimming has lower hydration and sodium needs:

```
if segment = swim:
    hydration_rate = 0.45 L/hr (base for swim)
    sodium_rate = 500 mg/hr (medium sweater)
else:
    # Use standard run/bike calculations
```

Justification: Current edge function specifies swimming hydration at 0.35-0.55 L/hr (lower than running's 0.4-0.8 L/hr) due to water immersion and horizontal position.

Claude's Comments: Reality Check on Carb Recommendations

This section contains my analysis of whether the algorithm's carbohydrate recommendations are realistic and whether athletes might find them too high, too low, or appropriate.

Pre-Workout Carbs: Potentially HIGH for Beginners

Concern: The cumulative pre-workout totals may seem overwhelming to recreational athletes.

Persona	Total Pre-Workout Carbs (Workout A)	g/kg	Real-World Equivalent
Elena (52 kg)	161g	3.1 g/kg	~6 slices bread + 2 bananas + bowl oatmeal
Marcus (75 kg)	233g	3.1 g/kg	~9 slices bread + 3 bananas + large oatmeal
Sarah (82 kg)	255g	3.1 g/kg	~10 slices bread + 3 bananas + XL oatmeal

My Assessment:

- **Good news:** These totals (3.1 g/kg) fall within the research-supported 2-4 g/kg range discussed in the "Cumulative Pre-Workout Carbohydrate Considerations" section.

- **For elite/experienced athletes:** These numbers align with what competitive endurance athletes actually eat before long efforts. Marathoners and Ironman athletes routinely consume 2-3 g/kg in the hours before racing.
- **For beginners like Sarah:** 255g of pre-workout carbs may still feel intimidating and could cause GI distress if she hasn't practiced this volume. Recommendations:
  - Build up to these targets gradually over training cycles
  - Start with 2 g/kg total and increase based on tolerance
  - Not all three windows are mandatory—many athletes use only breakfast + small top-up

During-Exercise Carbs: Appropriate but Gut-Training Dependent

The numbers look reasonable for the respective gut training levels:

Gut Training	Our Cap	Research Range	Assessment
Low (Sarah)	40 g/hr	30-40 g/hr	✅ Appropriate - conservative for untrained
Moderate (Marcus)	60 g/hr	60 g/hr	✅ Appropriate - SGLT1 ceiling
High (Elena)	88-100 g/hr	90-120 g/hr	✅ Appropriate for trained gut with dual carbs

My Assessment:

- These align well with Jeukendrup's research (60 g/hr single transporter, 90-120 g/hr with glucose:fructose)
- Athletes with low gut training may still struggle with 40 g/hr and should start at 30 g/hr
- The intensity-based "slider" within the band is a reasonable MEALVANA decision, though not explicitly from research

Potential Athlete Pushback:

- **Workout C (4hr bike, Elena at 88g/hr = 350g total):** This is achievable for trained cyclists but requires multiple gels/bars per hour. Some athletes may find this volume hard to consume.
- **Workout B (45min easy, 2g/hr = 1g total):** Athletes may question why they need ANY carbs for a short easy run. Consider dropping to 0g recommendation for <60min easy efforts.

Post-Workout Carbs: Slightly LOW for Aggressive Recovery

Our recommendations:

Workout	Post-Workout Carbs	Research Recommendation
A (2hr)	1.0 g/kg	1.0-1.2 g/kg/hr for 4 hours
C (4hr)	1.2 g/kg	1.0-1.2 g/kg/hr for 4 hours

My Assessment:

- These are for the FIRST recovery meal/hour only, which is appropriate
- Research shows glycogen resynthesis is maximized with 1.0-1.2 g/kg/hr for the first 4 hours
- Athletes doing multiple sessions per day should continue eating 1.0-1.2 g/kg each hour
- Our single-meal targets are reasonable for typical training

Sodium: May Seem HIGH to Casual Athletes

During-exercise sodium for 2hr run:

Persona	Our Target	Common Sports Drink
Elena/Marcus (medium sweater)	749 mg/hr	~200-300 mg/hr (Gatorade)
Sarah (low sweater)	267 mg/hr	~200-300 mg/hr

My Assessment:

- Our sodium targets are HIGHER than what most sports drinks provide
- This is intentional—the blog cites research showing wide variation (200-2000 mg/L sweat concentration)
- Athletes accustomed to standard Gatorade may be surprised by 750 mg/hr recommendations
- **Recommendation:** Include guidance that athletes may need salt tabs or high-sodium drinks to hit these targets

Hydration: Reasonable but Weather-Dependent

During-exercise hydration for 2hr run at 22°C:

Persona	Our Target	Common Advice
Elena/Marcus	1013 mL/hr	400-800 mL/hr
Sarah	608 mL/hr	400-800 mL/hr

My Assessment:

- Our targets (75% of sweat rate) are at the HIGHER end of typical advice
- 1 liter/hour is achievable but aggressive—requires drinking every 6 minutes
- Some athletes may find this uncomfortable or impractical during races
- The calculation is evidence-based (75% sweat replacement) but practical limitations exist

Overall Verdict

Category	Assessment	Confidence
Pre-workout carbs	Potentially high for beginners	Medium - need user testing
During carbs	Well-calibrated to gut training	High
Post-workout carbs	Appropriate	High
Sodium	Higher than expected	High - but may surprise users
Hydration	Aggressive but evidence-based	Medium

Key Recommendations:

1. ✅ Cumulative pre-workout guidance added (recommend 2-4 g/kg total; see "Cumulative Pre-Workout Carbohydrate Considerations")
2. Consider 0g during-carb target for easy workouts <60 min
3. Include guidance about needing salt tabs to hit sodium targets

4. Add "beginner modifier" (0.8×) for athletes with low gut training in future versions
5. Include warnings when targets exceed practical consumption limits

Will Athletes Balk?

Yes, some will—particularly:

- Beginners seeing 200+ gram pre-workout totals
- Athletes expecting standard Gatorade to meet sodium needs
- Anyone unfamiliar with aggressive hydration (1L/hr)

Mitigation strategies:

- Frame targets as "goals to build toward" not immediate requirements
- Provide "beginner mode" with reduced targets
- Explain the research basis so athletes understand WHY targets are high
- Include food equivalents so numbers feel tangible